TryHackMe Writeup: Intro to LAN

Field	Details
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Program	DG Interns Hub, Week 1
Room	Intro to LAN
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Estimated time	15 min
Status	Completed (100%)

Objective

Learn about some of the technologies and designs that power private networks.

What I did, task by task

Network topologies

- Looked at how devices can be arranged: star, bus and ring, and also the idea of a mesh.
- Star is the most common today because one failed cable does not take the whole network down, but the central switch becomes a single point of failure.

Subnetting

- A big network is split into smaller subnets to improve organisation, performance and security.
- The subnet mask (or CIDR notation like /24) tells which part of an IP address is the network and which part is the host.

ARP

- ARP finds the MAC address that belongs to an IP address on the local network. The request is broadcast and the owner replies directly.
- It has no authentication, which is why ARP spoofing is possible.

DHCP

- DHCP hands out IP address, gateway and DNS settings automatically.
- The process is Discover, Offer, Request, Acknowledge (DORA).

Topologies compared

Topology	Advantage	Drawback
Star	Easy to manage, one device failing does not affect others	Central switch is a single point of failure
Bus	Cheap and simple	One cable fault can bring everything

		down
Ring	Orderly data flow	A break in the ring can affect the whole network

Key takeaways

- Network design affects both reliability and security.
- Subnets help to isolate systems, so one compromised segment does not expose everything.
- ARP and DHCP are useful but can be abused by someone inside the network.

Why this matters for a SOC role

A lot of internal attacks (ARP spoofing, rogue DHCP servers, lateral movement between subnets) rely on how a LAN works. Understanding it helps in reading network alerts and designing segmentation.

Proof of completion

